

CASE BACKGROUND

You are working as a **Data Analyst for the HR department** of a large organization.

The company is facing an employee-retention problem, and management wants to understand what patterns exist in its workforce data.

Your job is to investigate the data, discover meaningful patterns, and communicate what those patterns could mean for the business.

No machine learning or predictive modeling is required.

All findings should be supported by appropriate **statistics, visualizations, and evidence from the dataset.**

SECTION A — EXPLORATORY DATA ANALYSIS

Q1 — Where Is the Attrition Problem?

10 Marks

The HR manager believes that employee attrition may be concentrated in particular parts of the organization.

Using the dataset, investigate **Department**, **JobRole**, and **OverTime** in relation to **Attrition**.

Identify the groups with noticeably different attrition rates and use appropriate visualizations and numerical evidence to explain **where HR should investigate the attrition problem first.**

Q2 — Are Employees Leaving Because of Their Work Experience?

10 Marks

HR suspects that employees' workplace experience may be related to their decision to leave.

Analyze **JobSatisfaction**, **EnvironmentSatisfaction**, **JobInvolvement**, and **WorkLifeBalance** in relation to **Attrition**.

Determine what patterns exist and explain **which aspect of employee experience appears most relevant for further HR investigation**, supporting your explanation with data.

Note: Do not claim that these factors directly *cause* attrition. Base your conclusions on the patterns observed in the dataset.

Q3 — Who Is Leaving the Organization?

10 Marks

Senior management wants to understand whether employees who leave are different from employees who stay.

Compare employees who **Stayed** and **Left** using variables such as **Age**, **MonthlyIncome**, **TotalWorkingYears**, **YearsAtCompany**, and **YearsInCurrentRole**.

Use suitable statistics and visualizations to identify meaningful differences and explain **what the data tells HR about the characteristics of employees who are leaving**.

SECTION B — FEATURE ENGINEERING

Q4 — Create a Career Stage

10 Marks

HR currently looks at employee age and experience separately, making it difficult to understand an employee's overall **career stage**.

Create a new **CareerStage** feature using appropriate existing variables such as **Age**, **TotalWorkingYears**, or **YearsAtCompany**.

Then use your new feature to investigate **whether attrition differs across career stages**.

Explain why your engineered feature is useful from an HR perspective.

Q5 — Build an Employee Experience Indicator

10 Marks

HR has several separate measures of employee experience, including:

- JobSatisfaction
- EnvironmentSatisfaction
- JobInvolvement
- WorkLifeBalance

Management wants a simpler indicator that summarizes an employee's overall workplace experience.

Create your own **EmployeeExperienceScore / indicator** using at least **three relevant variables**.

Use the new feature to investigate its relationship with **Attrition** and explain **what additional insight this engineered feature gives HR.**